Architectural Particular Specification

IRONMONGERY

SECTION 7.3

Rev	Date	Description
0	14 APR 2022	First Issue
A	13 MAY 2022	Tender Addendum No.2

SECTION 7.3

IRONMONGERY

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. The Work of this Section includes, but is not limited to, the furnishing and installing of the following:

1. All specified ironmongery and required fixings.
2. Ironmongery adjustments.
3. Fire testing and assessment of ironmongery on fire rated door assemblies.
4. Furnishing Ironmongery Schedule and templates to related trade for preparation of door hardwares.

1.02 REFERENCES

A. British / European Standards:

1. BS 476: Section 31.1: 1983 (1994) - Method of Measuring Smoke Penetration Through Doorsets and Shutter Assemblies Under Ambient Temperature Conditions.
2. BS 476: Part 23: 1987 - Method of Determination of the Contribution of Components to the Fire Resistance of A Structure.
3. BS 3621: 1980 (1992) - Specification for Theft Resistant Locks.
4. BS 5466: All parts - Methods of Corrosion Testing of Metallic Coating (date varies).
5. BS 5499: Part 1: 1990 (1993) - Specification for Fire Safety Signs.
6. BS 8220: Part 2: 1987 - Guide to Security of Building Against Crime: Offices and Shops.
7. PD 6512: Part 1: 1985 - Use of Elements of Structural Fire Protection With Particular Reference to the Recommendations Given In BS 5588: Guide to Fire Doors.
8. EN1125 Panic exit devices
9. EN1154 Door closing devices
10. EN1158 Door coordinators
11. EN1634-1 Fire testing
12. EN12209 Locks and Latches

B. Hong Kong Standards:

1. Design Manual Barrier Free Access 2008
2. Code of Practice for Fire Safety in Buildings 2011.

C. American Standards

1. ANSI A115: Specification for Door and Frame Preparation.
2. ANSI/BHMA A156.1 - 1988 : Butts and Hinges
3. ANSI/BHMA A156.2 - 1989 : Bored and Preassembled Locks & Latches

4. ANSI/BHMA A156.3 - 1994 : Exit Devices
5. ANSI/BHMA A156.4 - 1992 : Door Controls - Closers
6. ANSI/BHMA A156.13 - 1994: Standard for Mortise Locks and Latches
7. ANSI A117.1-1986: Specifications for Making Buildings and Facilities Accessible to, and Usable By, the Physically Handicapped.
8. ASTM E283-94: Test Methods for Rate of Air Leakage through Exterior Windows, Curtain wall and Doors.
9. ASTM E152-81a: Methods of Fire Tests of Door Assemblies.
10. NFPA 80 - Standard for Fire Doors and Fire Windows, 1992.
11. NFPA 101 - Life Safety Code, 1994.
12. NFPA 105 : Smoke Control Door Assemblies, 1993
13. NFPA 252 - Standard Method of Fire Tests of Door Assemblies, 1990.
14. National Institute of Law Enforcement and Criminal Justice (NILECJ): Standard 0306.00-Physical Security of Door Assemblies and Comments.
15. ANSI/ASTM F476-84: Standard Test Methods for Security of Swinging Door Assemblies.

1.03 QUALITY ASSURANCE

A. The ironmongery package shall be completed under one supplier, whom should have been in the local industry for over 15 years with recognized and approved experience.

B. The supplier's permanent, qualified staff with over 15 years of recognized local experience in complicated multi-million ironmongery supply contract, shall provide full local technical support;

C. Key management staff from the suppliers shall be a Registered Architectural Ironmonger from the Institute of Architectural Ironmongers, UK or a Door Hardware Consultant from the Door & Hardware Institution USA;

D. The supplier shall have experience in preparation of architectural hardware specification, estimating, detailing, master-key planning, ordering, servicing and supervising the installation of hardware and co-ordination from other trades; and

E. The supplier shall be responsible for coordinating with the Main Contractor and fitting out Subcontractors as well as door suppliers on rated or non-rated metal or timber doors for delivery and installation of ironmongery, which are appropriate for their intended use and fit its designated locations.

F. All locksets and ironmongery shall be supplied complete with all accessories such as escutcheons, fixing screws, etc. The accessories shall be compatible with, and matching in color, texture and material to the visible finish of the associated items of ironmongery.

G. Fixing screws shall be stainless steel of correct diameter to suit holes in the ironmongery to be installed. They shall be countersunk, half round or round-headed as required and with Philips head if visible. On no account are chromium or brass plated screws to be used.

H. Unless otherwise specified, Contractor shall supply and install architectural ironmongery to comply General Specifications issued by Architectural Services Department.

I. Hardware shall be suitable and adapted for its required use and shall fit its designated location. Should any hardware as shown, specified or required fail to meet the intended requirements or require modification to suit or fit the designated location, determine the correction or modification necessary and notify the Architect in ample time to avoid delay in the manufacture and delivery of hardware. The Contractor shall be deemed to have allowed for all such modification in the Contract Sum.

J. Qualification of Ironmongery and Fire/Smoke Seal Manufacturers: Companies specializing in the manufacturer of commercial grade, heavy duty ironmongery and seals (which have been fire tested and certified by BRE or other approved testing agencies and accepted by BD) with minimum of ten years documented experience.

K. Qualification of Ironmongery and Fire/Smoke Seal Installer :

Company specializing in the installation of fire-rated door assemblies and ironmongery with minimum of five years documented experience.

L. Field Service: The Ironmongery Supplier shall assign a competent representative, acceptable to the Architect, to be at the Project Site each time a major shipment of finish ironmongery is received. Such representative shall assist in "checking in" these shipments and shall secure a receipt covering the contents of each shipment. In addition, such representative shall be available for immediate call to the Project Site when, in the opinion of the Architect, his presence is necessary.

1.04 PERFORMANCE REQUIREMENTS

A. Security:

1. All ironmongery used and installed on the Project Building and Site shall be High Security Class IV/Grade 40 for all external doors and Class III/Grade 30 for all interior doors as conforming to ANSI/ASTM F476-84 standard.

2. The security and performance standards for Class III/Grade 30 and Class IV/Grade 40 requirements are indicated in Table 1 below :

Table 1: Ironmongery Performance Requirements

ITEM	DOOR ASSEMBLY TESTS	COMPONENT TESTS	MEASURED PARAMETER	REQUIREMENT[1]	
				CLASS III	CLASS IV
A.	Bolt Projection Strike Hole	Lock	Projection Size	$^{11}/_{16}$ in. (17.5 mm)	$^{11}/_{16}$ in. (17.5 mm)
B	Bolt Pressure	Lock	Resistance	150 lbs (670 N)	150 lbs (670 N)
C.	Jamb/Wall Stiffness	Jamb/wall	Force to spread Increase in lock-front to strike space	3,600 lbs (16,000 N) $^{1}/_{2}$ in. (13 mm)	4,950 lbs (22,000 N) $^{1}/_{2}$ in. (13 mm)
D.	Knob Impact[3]	Lock	Impact resistance	Five blows of 74 ft-lbs (100J)	Ten blows of 74 ft-lbs (100J)
E.	Cylinder Core Tension	Lock	Resistance	2,470 lbs (11,000 N)	2,470 lbs (11,000 N)
F.	Cylinder Body Tension	Lock	Resistance		3,600 lbs (16,000 N)
G.	Knob Torque[3]	Lock	Resistance	81 lbs-ft (110 Nm)	118 lbs-ft (160 Nm)
H.	Cylinder Torque[4]	Lock	Resistance	81 lbs-ft (110 Nm)	118 lbs-ft (160 Nm)
I.	Cylinder Impact[4]	Lock	Impact Resistance	Five blows of 74 ft-lbs (100J)	Ten blows of 74 ft-lbs (100J)
J.	Door Impact	Door	Impact resistance at center of panel	Six blows of 118 ft-lbs (160	Eight blows of 148 ft-lbs (200

				J)	J)
K.	Door Glazing Impact	Glazed panel installed in door	Impact resistance at centre of glazed panel	Five blows of 74 ft-lbs (100J)	Ten blows of 74 ft-lbs (100J)
L.	Hinge Pin Removal[5]	Hinge	Resistance	200 lbs (900 N)	200 lbs (900 N)
M.	Hinge Impact	Door, hinge, jamb/wall	Impact resistance at hinge	Six blows of 118 ft-lbs (160 J)	Eight blows of 148 ft-lbs (200 J)
N.	Bolt Impact	Lock, door, jamb/strike	Impact resistance at hinge	Six blows of 118 ft-lbs (160 J)	Eight blows of 148 ft-lbs (200 J)

Table 1 Notes :

1. These requirements promulgated by the National Institute Law Enforcement and Criminal Justice (NILECJ) Standard 0306.00 are incorporated in ANSI/ASTM 476-84 Standard Test Methods for Security of Swinging Door Assemblies.
2. Dead latch plunger must not enter strike hole with latch bolt.
3. Applies to Type A locks only. Type A locks use a single dead latch or the combination of a latch or dead latch and a dead bolt which are mechanically interconnected. Type B locks have a latch or dead latch mechanically independent from the dead bolt.
4. Applies to out-swinging doors only. All hinges of exterior doors and doors in high security areas shall have non-removable pins (NRP).
5. All locks shall provide a minimum one thousand effective key variations to BS 8220: Part 2 and shall be mortice locks, unless noted otherwise.
6. An electrical Master Key System including electronic strikes shall be provided for installation in the factory with the door sets.
7. External doors and high risk/security areas shall be keyed individually, unless otherwise noted.

B. Ironmongery Schedule:

1. The lengths of the cylinders are to be such that the cylinders protrude from the door by the minimum amount. The ironmongery supplier is to submit shop drawings indicating the proposed arrangement for approval, prior to ordering.

2. Refer to ironmongery schedule for types and function, both door and exterior security shutters.

C. Fire & Integrity:

1. All ironmongery, ancillary components and hardware on all fire rated doors and door frames shall be non-combustible with minimum melting point temperature of 800°C.

2. All ironmongery shall have been fire tested and SISIR certified in conjunction with the fire door and door frame as an integral assembly. No essential, non-sacrificial ironmongery and ancillary components and hardwares on fire rated doors and door frames shall fail, or become in-operative, during the fire-rated resistance period of the fire door/assembly specified.

3. All ironmongery and ancillary components and hardwares shall be tested in strict accordance with BS 5872, BS 6459 and BS 7352 for durability and impact resistance.

4. All door and door frame intumescent firestop and smoke barrier seals shall be “CERTIFIRE” and BBA approved, or equivalent UL-labeled seals. All door smoke seals shall comply with the requirements of Clause 1.03A of Section 08110.

5. Maximum width of lock mortice shall be 18mm maximum for 44mm thick fire rated door. Allowable mortice width shall meet fire test requirement and fire rating of door.

D. Handicap Accessibility:

1. Door closing devices (including door closers, spring hinges and floor hinges etc.) shall be designed to allow exterior and interior doors to be opened with forces of not more than 30 N and 22N respectively. Closers for interior doors shall have a closing period of at least 3 seconds measured from an open position of 70o to a point 75mm from the closed position measured from the leading edge of the door.
2. Door ironmongery shall comply with ANSI 117.1-1986 (Specifications for Making Buildings and Facilities Accessible to, and Usable By, the Physically Handicapped) and shall be easy to grasp with one hand without requiring grasping, pinching, or twisting of wrist to operate door latch.

1.05 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. The Contractor and Manufacturers shall be aware of the psychometric parameters used by the Consulting Mechanical Engineer in determining indoor temperature and relative humidity requirements. Indoor air quality requirements and purging procedures.

B. The Contractor and the Products Manufacturers shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without electrical circuitry and mechanism failures, corrosion, deformation and material failure or defects.

1.06 SUBMITTALS

A. Submit the following under provisions of Section 1.5.

B. Mill, fire acoustical, and test certificates for each type of ironmongery, fasteners, hardware components and accessories. BRE fire test certificates for all intumescent, acoustical and smoke seals.

C. Detailed (door by door) ironmongery schedule together with samples of all ironmongery items for approval by the Architect prior to order. All components shall be identified by Manufacturer's name and reference number and cover the requirements of smoke stop and intumescent seals on fire rated doors, detail door thickness, handing, suiting and master keying. Upon approval, this schedule shall form the basis for confirming the Contract requirements for ironmongery. The schedule shall be in a format approved by the Architect. Sample ironmongery shall be supplied in the finish specified.

The items indicated in the Ironmongery Schedule in Part 2 of this section are a general indication of the Project ironmongery requirement. It is the contractor's responsibility to develop this schedule into a full detailed ironmongery schedule.

D. Manufacturer's Statement of Review as required in Section 01 78 00.

E. Submit samples of all door and frame intumescent, acoustical, and smoke seals and weatherstrippings for the Architect's Review.

F. Submit Manufacturer's ironmongery and hardware operation, maintenance and lubrication information and schedule in English and Chinese.

G. The Ironmongery Supplier shall submit delivery schedule of all ironmongery to the Contractor for his preparation of the Programme of Work as required in Section 01 32 00.

1.07 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.08 DELIVERY, STORAGE AND HANDLING

A. Ironmongery Manufacturers shall deliver to the Ironmongery Installer ironmongery and ancillary hardware in original individual containers, complete with necessary appurtenances including installation tools, fasteners and instructions. The Ironmongery Installer shall assemble and repackage all ironmongery, hardware and seals into Ironmongery Sets corresponding to the approved Ironmongery Schedule. Each ironmongery set package shall be marked with the appropriate set number as indicated in the Ironmongery Schedule.

B. Accept ironmongery, hardware, seals and weather stripping on Project Site and inspect for damage, and missing components. Take inventory of all delivered ironmongery, hardwares and seals. Ensure that all ironmongery and seals are genuine products of the specified Manufacturers.

C. The Contractor shall be responsible for protecting all ironmongery, hardware, seals and weatherstrip materials against theft and damage (physical & climatic) during storage.

D. Thumbturns, handles, pulls and other items of finish ironmongery with easily damaged finishes shall be individually wrapped before placing in containers and with sufficient sheet cloth or cotton-backed paper which shall be adequately tied with heavy strings; all as necessary to protect the finishes.

E. Finish ironmongery shall be delivered, as directed, to the Project Site or the shops of the various fabricators to which such hardware is to be applied. Deliver ironmongery in the order required and in ample time to permit application at the Project or fabricator's shops, within the time required for the completion of the Project Building.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8 .03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

(NOT USED)

1.11 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. The Contractor, Product Manufacturers and the Ironmongery Installer shall jointly provide a five (5) year warranty against material defects, improper functioning, lock mechanism failures, and poor workmanship for Work of this Section commencing from the date of Project Practical Completion.

C. Material defects shall include, but not be limited to, electrical circuitry and mechanism failure, breakage, corrosion, lever sagging, fluid/lubricant leakage and other defects during the warranty period.

1.12 SPARE PARTS

A. This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

B. Provide Employer with 5% each of the total typical door ironmongery, smoke seals, and weatherstrips furnished, subject to a minimum of 1 item and a maximum of 10 items being provided.

C. Quantities of spare door ironmongery, fire/smoke seals, and weather strippings shall be proposed by the Contractor for approval by the Architect.

PART 2 PRODUCTS

2.01 MATERIALS

A. Refer to drawing no. MC-CA-501 to MC-CA-502 for Ironmongery Schedules and specification in this section.

B. Door Hinge

Spec Ref :	H1
Product Description :	IVES 5BB1HW 4.5x4.5 NRP 630 SS316 4.5"x4.5"x4.6mm 4 ball bearing heavy weight full mortise hinge with non removable pin, pack with screws for wood and metal door, tested to ANSI/BHMA A156.1 grade 1, UL listed, tested to EN1634, grade 316 stainless steel, satin finish
Source :	The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Spec Ref : H2

Product Description : IVES 5BB1HW 4.5x4.5 NRP 630 TW8M SS316
4.5"x4.5"x4.6mm 4 ball bearing heavy weight conductor hinge with 8 wires & reed switch door contact, with non removable pin, pack with screws for wood and metal door, tested to ANSI/BHMA A156.1 grade 1, UL listed, tested to EN1634, grade 316 stainless steel, satin finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

C. Door Closer

Spec Ref : C1

Product Description : LCN 4040XP Rw/PA 689

Heavy duty overhead door closer with regular arm and parallel arm bracket, cast iron body, adjustable size 1-6, adjustable closing speed and latch speed, with adjustable backcheck, tested to ANSI A156.4 grade 1 , UL listed, tested to EN1154 and CE marked, tested to EN1634, sliver finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

LCN

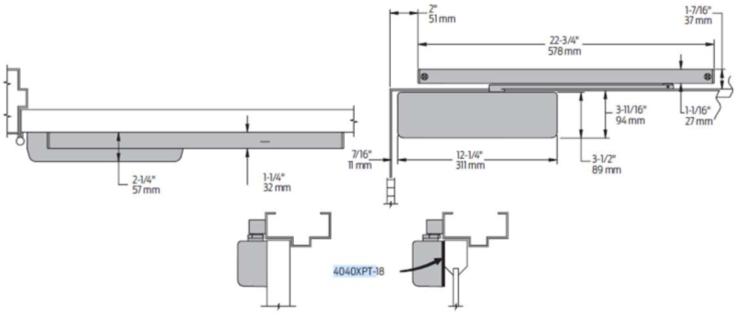
Spec Ref : C2

Product Description : LCN 4040XPT 689

Surface mounted heavy duty overhead door closer; sliding track arm, silver finish. Double heat-treated pinions gives proper hardness and strength to the parts. Liquid X fluid performs in temperatures from -30°F to 120°F, FAST power adjust for accurate on-site spring power adjustment, cast Iron construction offers proven durability, 30years warranty. Adjustable Spring Size 1-4, tested to ANSI A156.4 grade 1, UL listed, EN1634 2 hrs fire rated, 100 hour salt spray, ADA accessibility guidelines, 10,000,000 cycles tested, cast Iron Body material

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

LCN®



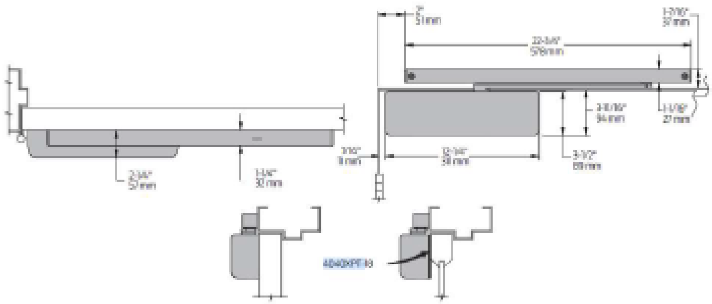
Spec Ref : C3

Product Description : LCN 4040XPT-H 689

Surface mounted heavy duty overhead door closer; sliding track arm with mechincal hold open finish, silver finish, Double heat-treated pinions gives proper hardness and strength to the parts. Liquid X fluid performs in temperatures from -30 F to 120 F, FAST power adjust for accurate on-site spring power adjustment, , Cast Iron construction offers proven durability, 30years warranty. Adjustable Spring Size 1-4, tested to ANSI A156.4 grade 1, UL listed, EN1634 2 hrs fire rated, 100 hour salt spray, ADA accessibility guidelines, 10,000,000 cycles tested, cast Iron Body material.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

LCN



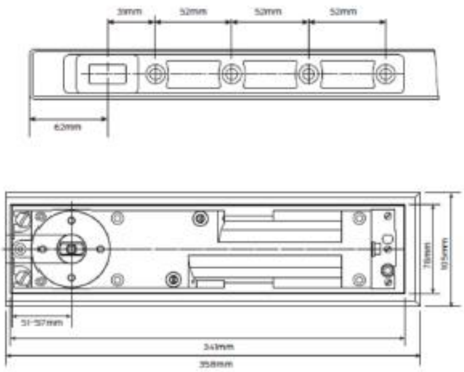
Spec Ref : C4

Product Description : Briton 7500 SSS

Heavy duty floor spring; size EN 3-6 (door width up to 1400mm), non-handed, for door weight up to 300kg, non hold open with stainless steel cover plate, in SSS finish, cast iron closer cylinder construction, adjustment concealed screws prevent tampering, tested to 500,000 cycles. Includes strap, top centre and satin s/steel cover Satin stainless steel finish; Warranty 10 year

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Briton



Spec Ref : C5

Product Description : BEA EYETECH

Infrared safety sensor strips, to be mounted on top of each sides of door, suitable for use in accordance with IP54, air humidity less than 95%, non-condensing, conforms to EN 61000-6-1, EN 61000-6-2, EN 61000-3 & EN 61000-4.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Spec Ref: **C6**

Product Description :

SESAMO PROSWING-F

Automatic swing door operator for double leaf door EN4-EN6, tested to BSEN1634-1, with standard arm set for push out side installation, for leaf weights up to 250 kg with door widths of 950 mm, safe opening and closing under adverse conditions. Meets 3.000.000 cycles.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Pushing articulated arm

TECHNICAL SPECIFICATIONS

POWER SUPPLY 230 Vac - 50HZ
NOMINAL POWER 85 W
POWER EXTERNAL DEVICES 12Vdc - 12W max
EMERGENCY BATTERIES 2x12 V 1.3 Ah (optional only for PROSWING M)
PARAMETERS ADJUSTMENT With double display with push buttons on control board, Dip-Switches or PC Interface
OPENING MAX. ANGLE 110°
MAX PAYLOAD 250 Kg (see chart above)
WING DIMENSIONS from 700 mm to 1400 mm
OPERATING TEMPERATURE -10°C a +50°C
WEIGHT = 12 kg
DIMENSIONS 664 x 130 x 86 mm
SERVICE Continuous
ELECTRIC LOCK POWER 12/24 Vdc - 15W max
SPRING TENSION EN4÷EN6

Arm system:

- ARTICULATED PUSHING ARM
- RIGID PULLING ARM

D. Locks, Latches and Bolts

Spec Ref : L1

Product Description :

Schlage L9092L-RX-03A-US32D +30-138 626

Electrified mortise lock with lever handle, outside lever controlled (fail safe/fail secure field reversible), inside lever always free to exit, outside key override, 70mm backset, 19mm throw stainless steel latch with anti friction tongue, request to exit switch, 12/24V DC input, ANSI A156.13 grade 1, UL listed for 3 hour fire door tested to EN1634, satin stainless steel finish, with Full size interchangeable core(FSIC) mortise cylinder housing and core for Schlage L series lock, with compression ring, spring & blocking ring, 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Spec Ref : L2

Product Description : Schlage M80ES-RL-03B-630 +30-138 626

Electro mechanical lock with lever handle set, 70m backset, 19mm throw stainless steel latchbolt with anti friction tongue, with auxiliary latch deadlock latchbolt when door is closed, both side lever controlled, outside lever fail secure, inside lever fail safe, with latchbolt monitoring switch and request to exit switch, tested to ANSI A156.13 grade 1, UL listed for 3 hour fire door, tested to EN1634, 12/24VDC input, satin stainless steel finish. With Full size interchangeable core (FSIC) mortise cylinder housing and core, 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Spec Ref : L3

Product Description : Schlage L9456 03A 630 +30-138 626

Office function mortise lock with lever handle, with outside cylinder & inside thumbturn, with deadbolt monitoring switch to output signal to overhead lamp indicator, 70mm backset, 19mm throw stainless steel latch with anti friction tongue, ANSI A156.13 grade 1, UL listed for 3 hours fire door, tested to EN1634 2 hours for timber door, satin stainless steel finish, with Full size interchangeable core (FSIC) mortise cylinder housing and core, 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

SCHLAGE

- Latchbolt retracted by lever/ knob from either side
- Deadbolt actuation by key or thumbturn rotation
- Throwing deadbolt locks outside lever/knob
- Turning inside knob/lever retracts both deadbolt and latchbolt and unlocks outside lever/knob
- Inside lever always free for immediate egress

SPECIFICATION : Refer to particular specification for full details.

Spec Ref : L4

Product Description : Schlage L9080L 03A 630 +30-138 626

Storeroom function mortise lock with lever handle set, 70mm backset, 19mm throw stainless steel latch with anti friction tongue, ANSI A156.13 grade 1, UL listed for 3 hour fire door, tested to EN1634 2 hours for timber door, satin stainless steel finish, with Full size interchangeable core (FSIC) mortise cylinder housing and core, 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

L9080 F07
LV9080

Storeroom lock

Latchbolt retracted by key outside or by knob/lever inside. Outside knob/lever is always inoperative. Auxiliary latch deadlocks latchbolt when door is closed. Inside lever is always free for immediate egress.

Spec Ref : L5

Product Description :

Schlage L9070L 03A 630 +30-138 626

Classroom function mortise lock with lever handle set, 70mm backset, 19mm throw stainless steel latch with anti friction tongue, ANSI A156.13 grade 1, UL listed for 3 hour fire door, tested to EN1634 2 hours for timber door, satin stainless steel finish, with Full size interchangeable core (FSIC) mortise cylinder housing and core, 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Classroom lock

- Latchbolt retracted by lever/ knob from either side unless outside lever is locked by key
- Unlocked from outside by key
- Auxiliary latch deadlocks latchbolt when door is closed
- Inside lever always free for immediate egress

SPECIFICATION : Refer to particular specification for full details.

Spec Ref : L6

Product Description : Schlage L9044 03A 630 +09-611

Privacy function mortise lock with lever handle set, 70mm backset, 19mm throw stainless steel latch with anti friction tongue, ANSI A156.13 grade 1, UL listed for 3 hour fire door, tested to EN1634 2 hours for timber door, satin stainless steel finish, with coin turn outside and occupancy indicator, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

SCHLAGE

Hotel occupancy indicator
09-611
For lock function L9486P, this unit can be used with A or B roses. Requires a 1 3/8" (35 mm) cylinder for 1 3/4" (44 mm) doors. Specify finish when ordering separately.

L9044
LV9044
Privacy with coin turn outside

Latchbolt retracted by knob/lever from either side unless outside is locked by inside thumbturn or outside coin turn. Operating inside knob/lever, closing door, rotating inside thumbturn or rotating outside cointurn unlocks outside knob/lever. Specify per L283-056 for Torx® screws. Available with rose trim only. (Previously XL11-868)

Spec Ref : L7

Product Description : Schlage B250

Mortise deadlatches, 70mm backset, 19mm throw stainless steel latch, ANSI A156.36 grade 2, UL listed for 3 hour fire door, tested to EN1634 2 hours for timber door, satin stainless steel finish, with Full size interchangeable core (FSIC) and inside thumb-turn, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

SCHLAGE

Full size interchangeable core

Thumbturn for B250

Schlage	ANSI
B250	**E0122**

Single cylinder deadlatch

- Deadlatch retracted by key outside or thumbturn inside.
- Holdback feature standard.

Outside **Inside**

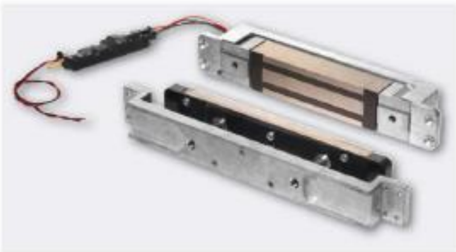
Spec Ref : L8

Product Description : Schlage GF3000 DSM/MBS

Electromagnetic shear lock with 3000lbs holding force, with built-in automatic relock switch, adjustable time delay on relock, 0-30 seconds, with magnetic bond sensor and door status monitor, low temperature operation, meets ANSI A156.23 Grade 1 (one million cycles), tested to BSEN 1634-1 for 120mins fire rated door

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

SCHLAGE



Specifications

	GF3000
Holding force	3000 lbs
Input voltage*	12/24 VDC
Current draw	.90A @ 12VDC .45A @ 24VDC
Lock body Dimensions (L x H x D)	9 1/2" x 1 1/2" x 1 1/2"
Lock body with mounting tabs	11 9/16" x 1 1/2" x 1 1/2"
Threshold box	N/A
Armature	8 3/8" x 1 3/8" x 1/2"
Armature bracket	10 5/8" x 1 3/8" x 1"
Weight	7 lbs.
Certifications	UL 10C, cUL, CSFM, ANSI/BHMA 156.23

E. Push Bars

Spec Ref : PB1

Product Description : Von Duprin 98-L-F-ALK-3’-996L 630 + Schlage 30-138 626

Rim type fire exit device, with mechanical outside trim, with battery powered local alarm, tested to ANSI A156.3 grade 1, UL listed, tested to EN1634, satin stainless steel finish, with Full size interchangeable core(FSIC) mortise cylinder housing and core cylinder for local alarm and outside trim, , 6 pin, open keyway, stain chrome plated finish.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

VON DUPRIN®

Spec Ref : PB2

Product Description : Von Duprin RX-9827EO-F-ALK US32D

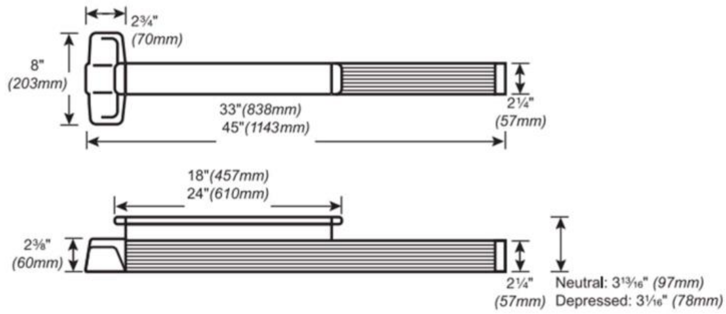
3' Surface vertical rod type fire exit device, for 711mm to 914mm door width with battery powered local alarm, request to exit switch, satin stainless steel finish.

Panel bar tested to ANSI A156.3 grade 1, UL listed, tested to EN1634 2 hrs fire rated, tested to 5,000,000 cycles

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

VON DUPRIN®

Dimensions



Spec Ref : PB3

Product Description : Von Duprin 1609 626
Strike plate for rim type painc device for double door application, satin chrome plated finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

VON DUPRIN®

Spec Ref : COR1

Product Description : IVES COR52 US28w/FL20
52" Bar type coordinator with 20" filler bar, for 62"to 92" double doors, tested to EN1158, UL listed, tested to EN1634, aluminum finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

IVES

F. Card Reader

Spec Ref : CR1

Product Description : Schlage MT15

Multi-technology reader with LED indicator and beep sound, contains 125 kHz proximity and 13.56 MHz contactless smart card in one-unit, shall support Schlage® Proximity, HID® Proximity, GE/CASI® Proximity, AWID® Proximity, LenelProx®, Schlage smart cards using MIFARE Classic®, MIFARE Plus®, MIFARE® DESFire® EV1 and EV2, DESFire® CSN, HID iCLASS® CSN, Inside Contactless PicoTag® CSN, ST Microelectronics® CSN, Texas Instruments Tag-It®, CSN, Phillips I-Code® CSN with 2.5-12.7cm card read range. Wiegand , Clock & Data and RS-485 (OSDP) output. Compatible with NFC-enabled mobile devices. Comply with ISO 14443A, 14443B and 15693 industrial standard and FIPS 201-1 & 201-2. UL294 listed, CE marked and IP65 rated. Black/cool-tone gray/ cream/ warm-tone brow finish available.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

G. Flush Bolts

Spec Ref :	FB1
Product Description :	IVES FB358 US26D Lever action manual flush bolt for timber doors, UL listed, tested to EN1634, satin chrome plated finish
Source :	The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

H. Dust Proof Socket

Spec Ref :	DP1
Product Description :	IVES DP1 US26D Dust proof strike, satin chrome plated finish
Source :	The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

DP1

I. Pull Handles

Spec Ref : PH1

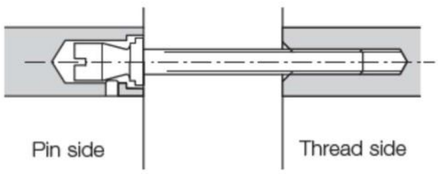
Product Description : ELITE PH201RS
Stainless steel pull handle, 19mm dia, 225mm centre to centre, with surface fixing on rose for timber or metal door, grade 316 stainless steel, satin finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

Spec Ref : PH2

Product Description : ELITE PH201BB
Stainless steel pull handle, 19mm dia, 225mm centre to centre, back to back fixing for timber or metal door, grade 316 stainless steel, satin finish, in pair

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754



Spec Ref : PH3

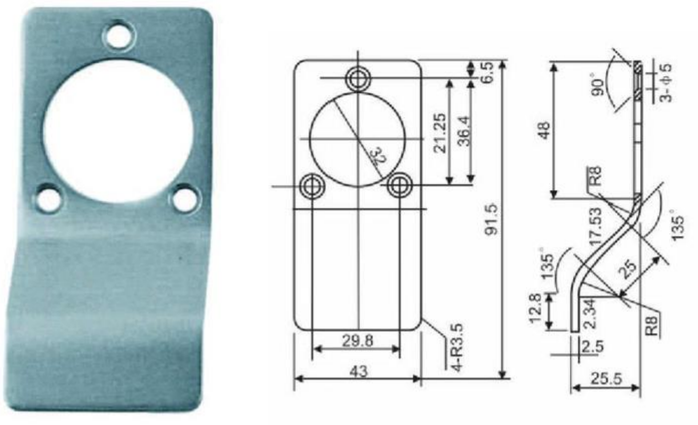
Product Description : ELITE PH600.32P
Stainless steel grab bar, dia.32mm x 600mm length, back to back fixing, 316 stainless steel, hairline finish, pair

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

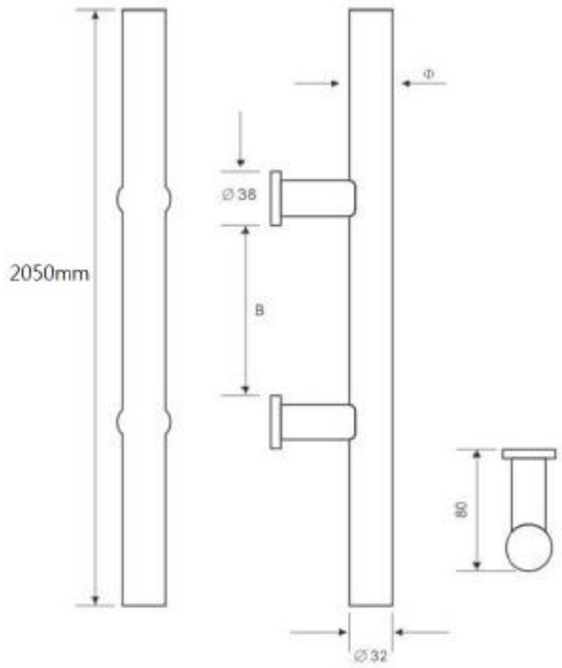
Spec Ref : PH4

Product Description : ELITE CP03
Stainless steel cylinder pull for round cylinder, satin finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754



Spec Ref :	PH5
Product Description :	ELITE PH2050P Stainless steel grab bar, dia.32mm x 2050mm length, back to back fixing, 316 stainless steel, hairline finish, pair
Source :	The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

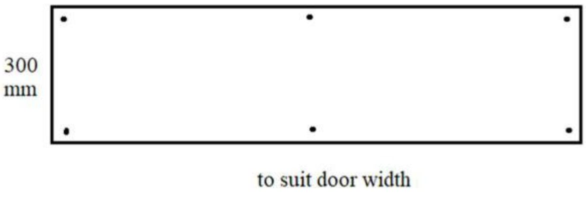


J. Kick Plate

Spec Ref : KP1

Product Description : ELITE KP01
Grade 316 stainless steel kick plate with radius corner and soften edge, surface fixing by countersunk screws, 300mm height x DOOR WIDTH, satin finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

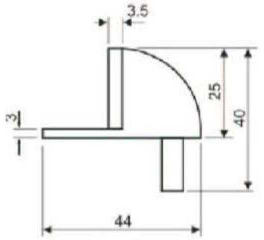


K. Door Stops

Spec Ref : DS1

Product Description : ELITE DS-01
Dome shape door stop, floor mount, 304 stainless steel, hairline finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754



L. Power Supply

Spec Ref : PS1

Product Description : Schlage PS904-4RL-FA
Power supply with 4 replay board and fire alarm relay board, 2A@12/24 VDC field selectable output, universal 120-240 VAC fused primary input.
UL294 listed.

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

SCHLAGE

M. Door Contact

Spec Ref : DP1

Product Description : Schlage 679-05
Concealed single pole double throw magnetic door contact, tested to EN1634

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754

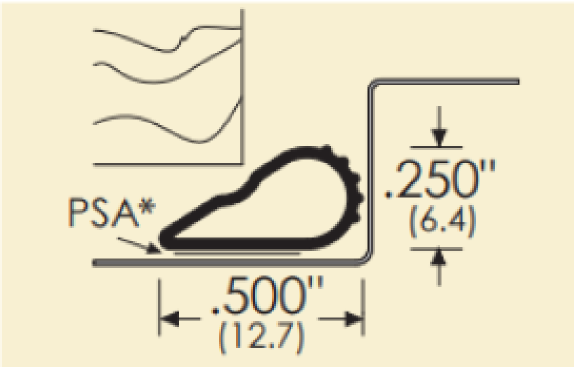
SCHLAGE

N. Door Seal

Spec Ref : S1

Product Description : Zero188S-Bk
Door seal, tested to EN1634, black finish(length to be confirmed)

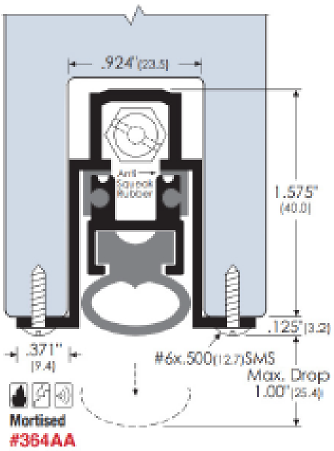
Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754



Spec Ref : ASB1

Product Description : Zero 364-AA
Auto door bottom, tested to EN1634, aluminum finish

Source : The Jardine Engineering Corporation Ltd. - Roy Lau 6392 1754



Door Hinges:

1. Hinges for timber doors shall be heavy duty weight, size 4.6(min.) x 114 x114mm, 4 ball bearings grade 316 stainless steel butt hinge of (H1) “***IVES***” 5BB1HW 4.5x4.5 NRP 630 SS316, with stainless steel machine screws or equivalent.
2. Hinges shall be full mortise type; unless otherwise specified with flush barrels; non-magnetic stainless steel pins, stainless steel washer, flat button tops and square corners.
3. Hinges for G.M.S. or stainless steel doors shall be heavy duty weight, size 4.6(min.) x 114 x 114mm, 4 ball bearings grade 304 stainless steel butt hinge. Hinges shall be (a) “***IVES***” 5BB1HW 4.5x4.5 NRP 630 SS316, with stainless steel metal screws or equivalent.
4. Hinge materials and finishes shall be selected to match other hardware.
5. Each leaf of door shall have 3 nos. of hinges. 4 nos. of hinge shall be provided to door leaf in size of exceed 2200mm (H), subject to door weight.
6. Provide full mortise Type 316 stainless steel hinges with stainless steel pins. Install non-removable pins (NRP) for all exterior conditions. Non-rising for non-security exposure, flat button with matching plugs. Install Type 316 S11 Austenitic Stainless Steel with 2.5% molybdenum door hinges for exterior doors and interior wet area locations.
7. Heavy-duty, ball-bearing type for all doors. Swaged, inner leaf beveled, square corners. Hinges shall comply ANSI A156.1 Grade 1, passed 2,500,000 cycling test.
8. A minimum of three hinges shall be installed per door leaf. Maximum permissible frictional torque measured at 4000 cycle of operations shall be 4 Nm. Maximum in-service torque shall be 5 Nm.
9. All hinges for fire-rated doors shall have a minimum melting point temperature of 800^{o}C.
10. Hinges with rising cams or quick action rising spindle may not be used in fire rated doors.
11. Install two hinge bolts per door leaf to provide protection against doors being forced open from the hinge side.
12. Electrified locks or similar to be provided with approved conductor hinges. Size 4.5” x 4.5” x 4.6mm 4 ball bearing heavy weight conductor hinge with 8 wires & reed switch door contact, with non-removable pin. Tested to ANSI A156.1 Grade 1, UL listed, Grade 316 stainless steel, stainless steel finish.

O. Locks, Latches & Bolts:

1. ANSI Grade 1 series mortise locks and be tested to 1,000,000 operations. ANSI A 156.13, Grade 1, UL Listed for 3 hour fire rating. A115.1 door preparation. Handing: field reversible
2. 2-3/4” Backset. Case: Stamped steel 0.090 thk., zinc dichromate finish. Front: Adjustable bevel, steel backplate, 0.110” thk. With Armor plate 0.060” thk., standard front: 1-1/4” x 8”. Deadbolt: stainless steel, full 1”throw. Hardened roll pins. Latch: stainless steel with steel anti-friction latch levers 3/4” throw, reversible. Auxiliary latch: 1/2” throw, no reversing necessary. Rocker switch: locks or unlocks hub. Lever support springs (2). Hub lock arm: steel 0.125 thk. Hubs: 9/32” for knob sets or 5/16” with lever support springs on square, sintered steel, with security separator.

3. Classroom lock. Latch by lever either side except when outside lever locked or unlocked by key. Auxiliary latch deadlocks latch. Inside always open. Inside lever retracts latch & deadbolt simultaneously.

4. Entrance lock. Latch by lever either side. Deadbolt by key outside, turnpiece inside. Deadbolt deadlocks latch.

5. Privacy lock. Latchbolt retracted by lever from either side. Deadbolt by outside key or inside thumbturn. Thrown deadbolt displays “OCCUPIED” and locks outside. Inside lever always free.

6. Cylinder Rim Nightlatch (w/ key alike system). Operates by key outside, turn knob inside. Night latch features thumb button used to hold back or deadlock bolt so that key will not operate with 5 pins tumbler security and a standard backset of 2-1/4” which makes it suitable for all new installations or as a replacement for existing night latches. Fits 1-1/2” to 2-3/4” thk. door reversible for either hand. Zinc alloy rim cylinder with 2 keys and high density, rust resistant alloy case (87 x 64 x 45mm).

7. Lever furniture shall be Grade 304 Satin Stainless Steel finish.

8. Rectangular wrought door pulls shall be Grade 304 Satin Stainless Steel finish, bolt-through or back to backfixing type to suit the operational requirements of each rooms.

9. Round wrought door pulls shall be Grade 304 and min. 19 mm dia. and 813c.1519/32D or equivalent.

10. Lever handle for all timber doors except sliding doors shall be “***Allegion***” Schlage ET511 or ET615 or equivalent.

11. Cylinder shall be Full size interchangeable core (FSIC) mortise cylinder allow cylinder re-key or replacement without disassembly of the lock. With housing and core, with compression ring, spring & blocking ring, 6 pin, stain chrome plated finish

12. Mortise Locks: ANSI A156.13, Operational Grade 1 for non-security doors and Security Grade 1 for exterior and security locations, equipped with 6-pin tumbler, “keyed differently”, unless noted otherwise. Provide 2-3/4 inch (70 mm) backset, unless noted otherwise. Wrought-box strikes. Latch bolt minimum throws indicated in Ironmongery Schedule.

13. Latch Sets: Provide push-button releases by turning lever, closing door, or turning emergency release key through hole in outside knob.

14. Strikes: ANSI Strikes, 1-1/4 x 4-7/8 inches (32 mm x 127 mm), with curved lip. Wrought box strikes, with extended lip for latch bolts. Provide dustproof strikes for all foot bolts.

15. Tactile Warning: Provide locks with Manufacturer’s standard tactile warning per handicapped codes when required by local Governing Authority.

16. All door levers shall be manufactured with cast, solid-bar stock. Hollow, wrought fabricated levers with resin or metal filler are not accepted.

17. All forend plates to be finished, stainless steel hairline finish, unless noted otherwise.

18. Lock & Latch Testings

    a. Lock and Latch Testings:

    Three locks or latches shall be taken from consignment and tested in accordance with BS 5872, Appendix A. If any lock or latch fails, then a second test of three locks or latches shall be taken from the same consignment and tested in accordance with the same tests. If any lock or latch in the second test sample fails, the consignment shall be deemed not to comply with BS 5872 Standards. All failed consignment shall be replaced with new.

    b. The sequence of tests in accordance with BS 5872 shall be:

    (a) Freedom of operation of follower, knob, lever, and spring.

    (b) Strength of bolt spring, follower or cam, knob and lever.

    (c) Axial force on knobs or levels of bored or latch sets.

    (d) Strength of security mechanism of bored lock or latch set.

    (e) Strength of connecting bar in rim cylinders.

    (f) Dead bolt against end pressure.

19. Operation of security mechanism at 70 test. Operations per minute for 100,000 operations test. Any failure of the key or the lock or latch shall be deemed to be a failure of the test. No damage or distortion shall occur. Security Mechanism shall function properly after testing.

20. Proprietary high security locks together with proprietary electric strike lock system shall be provided at Staff Room and to be further confirmed by owner.

P. Door Closers:

1. Overhead door closers shall conform to ANSI A156.4 grade 1 or BS EN1154 : 1997 with Classification 4-8-2/4-1-1-3 OR 4-8-2/6-0-1-3 for closers with adjustable powers. Fixing shall be by template to ensure precise installation.

2. Adjustable devices shall be fully concealed and suitable adjustment tools shall be included if necessary.

3. Surface mounted closers shall be fitted internally on the side of the door where they are least visible to the public. They shall be of neat, aesthetically pleasing design and fully reversible. Cover plates shall not be of types which may be easily damaged or removed.

4. Closers for fire rated doors shall be ‘Certifire’ Approved to BS476 Part 22 or have been included in successful fire tests to BS 476 : Pt 22 : 1987 (and with the Certificate approved).

5. Overhead surface mounted door closers (w/ standard arm mounted) shall be variable closing force, size 2-4 to EN1154 door leaf width maximum to 1100mm guide rail only closing force EN3 (950mm door width). Adjustable latching action accelerates over the last 10 degree of closed position, approximately in order to overcome latch resistance, door seals or wind pressure etc.

6. Heavy duty overhead surface mounted door closers (w/ standard arm mounted /parallel arm bracket) certified with ANSI A117.1 Barrier Free and ANSI A156,4 Adjustable Delay Action and Back-Check function for elderly or handicapped, UL Listed Grade 1 requirement. Finish: Silver. Loading force: EN2-6

7. ANSI A 156.4, Grade 1. (teste to 2 million cycles) Leak-proof gasketed fluid housing.

8. ADA/ANSI A117.1 and U.L. listed for Class A.

9. To comply with Design manual Barrier Free Access 2008,

10. Non-Handed, Non-Sized; adjustable 2-4. 180 degree door opening.
11. Heavy duty parallel arm of solid forged arms equipped with saw-resistant steel roll pins in the arm studs to prevent disassembly.
12. Cover shall be Stainless Steel Type 316 S11 with four point attachment, concealed security screws.
13. Multiple backcheck location valve.
14. Extreme temperature hydraulic fluid, non-combustible.
15. Delayed action closing, 15 seconds delay.
16. No through bolting allowed.
17. Provide exposed metal cover to match hardware.
18. Size and mount units indicated or, if not indicated, to comply with Manufacturer's recommendations for exposure condition. Reinforce substrate as recommended.
19. Consignment of door closers shall be independently tested to 10 million cycles. Throughout the test, no fluid leakage, or damage to door closer and arm shall occur.
20. Closers shall not be visible on the public side of doors, unless noted otherwise. Closers opening into public spaces shall be provided with parallel arms and brackets to suit.
21. Closers shall be sized in accordance with the accepted Manufacturer's standards to suit height, width, weight of door, wind/draft conditions, and staircase pressurization.
22. Fixed floor spring, size 4, for doors weight up to 100kg weight, cast iron closer cylinder construction, concealed screws prevent tampering, tested to EN1634 and 500,000 cycles. With strap, top center and cover, satin stainless steel finish.
23. Provide a top pivot for each floor closer. Provide weather sealing compound for each exterior floor closer.

Q. Panic Exit Devices and Security Alarm System

1. Emergency exit devices shall conform to the minimum requirements of ANSI A156.3-1994 Grade 1 standards. Rim or mortise type with smooth mechanism case, satin stainless steel with UL durability cycle tested to 10,000,000 cycles.
2. Also comply to BS EN 1125:2008 and CE Marked
3. For public entrance of circulation doors, such devices shall have key access from outside.
4. For doors where access is required from the outside, they shall have key operated locking attachments.
5. In locations where security is paramount they shall incorporate local alarms fitted to the vertical bar.
6. Exit alarm locks shall be bold, intimidating design to deter misuse. Cast aluminum body to protect the internal mechanism and resists tempering or vandalism. Tested to withstand up to 1600 pounds of static load force against the door. 85-decibel alarm with armed indicator light and audible low battery alert signal. Stainless steel deadbolt of 51 x 13mm with 19mm projection. Perplex signage with "Emergency Exit Only – Alarm Will Sound" sign built into the device and cover the full width of push bar. Warning sign in English with Braille. The cross bar shall extend to the full width of the door for ease of escape, they shall comply with NFPA 101 Life Safety Code.

7. Keyed emergency exit devices shall be Full Size Construction Core, with masterkeyed together with other locks.
8. Mechanical panic exit devices shall accept an outside cylinder and/or lever handle operation.
9. Provide integral alarm with key reset, activate, or deactivate mode selection, remote monitoring and/or motor operation as required.
10. Electrical Push-bar shall be electrically controlled panic exit systems and interfaced to the electronic access control system. Electrical Push-bar shall be DC powered and fitted with local alarm.
11. Delay egress devices shall have an internal mechanism incorporated inside the panic bar providing a time-delayed panic exit of fifteen seconds. The locking stays secure for fifteen seconds on depression of the push rail. A key shall be provided to reset the system.
12. Delay egress devices shall release immediate on receipt of fire signal.
13. Delay egress devices shall be fitted with local alarm

R. Electrical Mortise Locks

1. ANSI Grade 1 series mortise locks and be tested to 1,000,000 operations. ANSI A 156.13, Grade 1, UL Listed for 3 hour fire rating. A115.1 door preparation. Handing: field reversible
2. 2-3/4" Backset. Case: Stamped steel 0.090 thk., zinc dichromate finish. Front: Adjustable bevel, steel backplate, 0.110" thk. With Armor plate 0.060" thk., standard front: 1-1/4" x 8". Deadbolt: stainless steel, full 1 "throw. Hardened roll pins. Latch: stainless steel with steel anti-friction latch levers 3/4" throw, reversible. Auxiliary latch: 1/2" throw, no reversing necessary. Rocker switch: locks or unlocks hub. Lever support springs (2). Hub lock arm: steel 0.125 thk. Hubs: 9/32" for knob sets or 5/16" with lever support springs on square, sintered steel, with security separator.
3. The lock shall be 12/24 VDC powered, auto detect, and shall be using different DC power source from electronic access control system controllers.
4. A Field selectable allow to configure the lock to fail-safe or fail-secure type to fulfill security design requirement. Monitor switches include request to exit as standard and optional deadbolt, latch bolt monitor and the following functional requirement:
   - Electrically locked (fail safe) : Outside knob/lever continuously locked by DC power. Latchbolt retracted by key outside or by knob/lever inside. Switch or power failure allows outside knob/ lever to retract latchbolt. Auxiliary latch deadlocks latchbolt when door is closed. Inside knob/lever always free for immediate exit. Inside lever is always free for immediate egress. Field reversible fail safe and fail secure feature. Key override.
   - Electrically unlocked (fail secure): Outside knob/lever unlocked by DC power. Latchbolt retracted by key outside or knob/lever inside. Auxiliary latch deadlocks latchbolt when door is closed. Inside lever is always free for immediate egress.
   - Double Motor : electrically locked on one side and unlocked on the other side (one-side fail safe and one-side fail secure) (i.e. critical areas)

5. Both inside and outside knob/lever unlocked by DC power. Latchbolt retracted by key outside. Auxiliary latch deadlocks latchbolt when door is closed. Field reversible fail safe and fail secure feature.
6. The door lock power shall be direct connected to automatic fire detection system for fail safe or fail secure operation. The exact requirement of each electrical mechanical mortise lock shall in conjunction with the door schedule, ironmongery schedule and security drawings for configuration.
7. Keyswitches shall accept euro single, modular cylinder of 30/10mm length for integration with the building master key suite, complete with SSS spanner tips tamper resistant screws. They shall be of grade 316 stainless steel for corrosion resistance.
8. Power sources for electric lock, keypads shall be UL Listed. If fire alarm systems do not incorporate clean contacts, transformers shall incorporate relay control interfaces.

S. Door Stops, Holders And Bumpers:

1. Door stop shall be manufactured in satin stainless steel of grade 304 and shall be DS674S/32D or equivalent.
2. Door stop mounting: Methods to suit substrates encountered (plastic anchor, plaster board anchor, expansion shield).
3. Provide grey rubber exposed resilient parts, unless noted otherwise.
4. For doors at the back of house, floor mounted type of door stop is preferred as specified in ironmongery schedule.
5. Adjust height of floor stops to suit undercut of adjacent door.

T. Ironmongery Lubricants:

1. Light oil for latches and bolts
2. Flake graphite or liquid lubricants containing PTFE (Polytetra floroethylene) particles in suspension for pin tumbler and disc tumbler cylinders.

U. Pivots:

1. Provide quantities and types of pivots (offset, intermediate and center) as required to suit door sizes and weights.
2. Pivot sets (offset and center) shall consist of top and bottom pivots, unless otherwise indicated.
3. Provide a top pivot for each floor closer, unless otherwise indicated.

V. Pushes And Pulls:

1. Provide concealed fasteners where practical. Where exposed fasteners are required provide flush type finished to match door push or pull.
2. Dimension 91.5 x 43mm. finger pull extends 25mm. fastened by three #6 x 5/8” tapping screws, attached by lock cylinder. Satin stainless steel grade 304.

W. Padlocks:

1. Padlocks shall be of appropriate grades to BS EN 12320, and capable of interchangeable Core (FSIC), master and construction keying together with other locks.

X. Flush Bolts:

Flush bolt and dust proof sockets for timber door shall be “IVES” FB358 US26D + DP001/26D or equivalent. Provide top and bottom extension type flush bolts, mounted 300mm and 1830mm, respectively, from the bottom of each door, where scheduled. Provide each bottom flush bolt with a dustproof strike. Satin stainless steel grade 304 and satin chrome. Meets ANSI A156.16 for L04201

Y. Silencers:

Provide silencers for all non-gasketed and non-weatherstripped frames. Provide four (4) for each single swing door and four (4) for each pair of doors.

Z. Thumbturn:

Stainless steel grade 316. Rose 52 diameter, with nylon bearing, stainless steel grade 316. All fixings concealed by a snap on cover. Thumbturn and rose assembly to have passed Class 4 of the draft EN standard.

AA. Push Plates

1. Plate shall be radius corner and softened edge and made by satin stainless steel of 1.5mm thick grade 304, with #6 x 5/8” tapping screw.

AB. Automatic Swing Operator

Automatic door operators shall be approved, high quality devices to BS7036, to suit the door construction, configuration, location and frequency of use and operate at 230/240vAC 50Hz. Finishes shall match other hardware on the door and units shall be frame mounted wherever possible, on the door face least susceptible to weather and/or tampering. Operators for fire rated doors shall be of suitable types which will not compromise the fire rating, linked to fire alarm systems to close automatically in case of fire. Operators shall allow adjustment of opening and closing speed, hold open time, backcheck and power and allow manual override, in case of power failure.

If one leaf of a pair of self-closing doors must close before the other, due to rebated styles, latch bolts, etc., supply suitable matching selectors to BS EN1158 Class 3-5-*-1-1-3 (where “*” denotes the appropriate door mass from Table 1), of types which do not obstruct ironmongery or affect fire ratings. Use sprung-arm, under-frame fixing selectors for outward opening doors and rebate or face fixing types for inward opening doors.

2.02 ANCILLARY MATERIAL : KEYS AND KEYING

A. The Contractor shall meet with the Client/SO to determine keying requirements and obtain final instruction before proceeding. Unless otherwise stated, keyed locks shall be master keyed under a system allowing the use of a Great Grand Master Key.

B. Construction Cylinders: Provide full size construction cylinders that are replaceable by permanent cylinders. Provide 8 construction master keys. Interchangeable cores allow cylinder rekey or replacement without disassembly of the lock.

C. Provide 3 sets of servant or differ keys for each lock and 3 sets of all other keys in the system. Keys to be nickel silver finish.

D. Manufacturer's direct representative shall fully instruct client's personnel to make system ready for operation. Keys are catalogued and installed from temporary key control cartons in to system by client and inspected by the manufacturer's representative periodically to assure cataloging and installation are done properly.

E. Keying shall be performed at factory. Each key shall be stamped with its Door Number (i.e.B1-23), then packed separately from the locks.

F. Master keys shall not be shipped with the shipment(s) of ironmongery but shall be delivered in strict accordance with instructions to be issued by the SO.

G. Deliver Master series type keys properly identified direct to client in sealed envelope.

H. Coordinate temporary Building Construction Master Keying requirements with the Architect and the Project Manager.

I. Where keys are required:

1. Provide twenty (20) key blanks for each master key system.
2. Provide three (3) Master Keys for each master key system.
3. Provide key control system, including key cabinet with capacity to store 150% of keys furnished for each system. The key cabinet in to be constructed in metal.
4. Final keying requirements will be determined by the Employer.
5. Master keyed system to be further confirmed with client's requirement.

J. Each set of keys shall be provided with an approved circular chromium plated brass plate 25mm diameter x 1.5mm thick stamped with the identification of room to which it belongs.

K. Properly arrange, identify, classify and tag any lock spanner wrenches, spare parts and any other tools furnished by the Manufacturers with the ironmongery for handing over to the Project Manager.

2.03 FABRICATION

A. Finish and Base Material Designations: Number indicates Builder's Hardware Manufacturers Association (BHMA) Code, or nearest traditional U.S. commercial finish.

2.04 IRONMONGERY & HARDWARE FINISHES

Nearest U.S. Equivalent	**BHMA Code**	**Finish Description**		**Base Material**
USP		600	Primed for	Steel
painting				Steel
US1B	601	Bright japanned		Steel
US2C		602	Cadmium	Steel
plated				Brass*
US2G	603	Zinc plated		Brass*
US3605		Bright brass, clear coated		Bronze*
US4606		Satin brass, clear coated		Bronze*
US9611		Bright bronze, clear coated		Bronze*
US10	612	Satin bronze, clear coated		Brass, Bronze*
US10B	613	Oxidized satin bronze, oil rubbed		Brass, Bronze*
US14	618	Bright nickel plated, clear coated		Brass, Bronze*
US15	619	Satin nickel plated, clear coated		Bronze*
US19	622	Flat black coated		
US20A		624	Dark oxidized,	Brass, Bronze*

Nearest U.S. Equivalent	BHMA Code	Finish Description	Base Material
statuary bronze, clear coated			Brass, Bronze*
US26	625	Bright chromium plated	Aluminium
US26D	626	Satin chromium plated	Aluminium
US27	627	Satin aluminium, clear anodized	Stainless steel 300 series
US28	628	Satin aluminium, clear anodized	Stainless steel 300 series
US32	629	Bright stainless steel	Brass, Bronze*
US32D	630	Satin stainless steel	Brass, Bronze*
---		684 Black chrome,	
bright			
---		685 Black chrome,	
satin			

Note
* Also applicable to other base metals under a different Builders' Hardware Manufacturers Association code number.

A. Where base material and quality of finish are not otherwise indicated, provide at least commercially recognized quality specified in applicable referenced standards in this Specifications.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that door and frames are ready to receive Work and dimensions are as instructed by the Manufacturer.

B. Verify that electricity is available to power all operated devices and of the correct characteristics and voltage.

3.02 INSTALLATION

A. Hardware Mounting Heights: Door and Hardware Institute Recommended Locations for Builder Hardware for Standard Steel Doors and Frames, except as otherwise indicated.

B. Install door ironmongery plumb, level and true to line, in accordance with Manufacturer's written instruction and recommendation. Install each ironmongery and hardware item on fire rated door and frame to comply with fire test certificates and requirements.

C. Lubricate all ironmongery and moving hardware as recommended by the Manufacturer.

D. External thresholds shall be set and bolt to concrete surfaces using lead expansion shields and countersunk stainless steel screws to match threshold color. Set threshold in bed of exterior sealant.

E. Install all door and frame intumescent and smoke seals, and weather strippings as indicated on the Ironmongery Schedule, and in strict conformance with fire test requirements.

F. Mortice work in correct locations and size, without gouging, splintering or causing irregularities in finished work. Any voids due to over mortising as a result of fitting ironmongery and hardware shall be neatly filled with suitable weatherproof intumescent paste.

G. Where cutting and fitting are required on substrates to be painted or stained, install, fit and adjust ironmongery prior to finishing. Then remove and reinstall door ironmongery after finishing. In situ finishing to doors without removal of ironmongery is not permitted.

H. ADJUSTMENT

Adjust door closers to compensate for the Building's Mechanical System, internal pressure and exterior wind conditions, so that a maximum force of 8.5 pounds (38N) can open exterior non-fire doors, and a maximum 5 pounds (22N) opening force for interior non-fire doors.

3.03 MAINTENANCE AGREEMENT

A. The Contractor shall return to Project Building one month after Employer's occupancy, and adjust hardware for proper operation and function. Instruct Project Building Maintenance personnel in proper maintenance and adjustment of all ironmongery at time of Project Handover.

B. Lubricate all ironmongery and moving hardware as recommended by the Manufacturers:

1. Sides and striking face of latch bolt shall be lubricated by a smear of light oil every 6 months. Pin tumbler and disc tumbler cylinders shall be lubricated with flake graphite or liquid lubricants containing PTFE particles in suspension.

2. Lubricate bolts with light machine oil at 6-month intervals.

C. All latches shall be inspected at 6-month intervals to ensure that latch bolt moves freely and projects fully at all times, especially when engaged.

D. Check and test exit devices and fire shutter release mechanism with Building Fire Protection and Life Safety System at least every two months.

E. Adjust door closers as noted in Clause 3.03A above.

F. Refer to door schedule for the requirements and location for weather stripping, thresholds, smoke seals, intumescent seals and drop seals, etc.

END OF SECTION